Capacity-Aware Fraud Triage via Conformal Routing and Shadow Pricing

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Abstract

We study fraud detection as a capacity-constrained human-AI collaboration problem. A bank must route 10,000 daily transactions across automated decisions, human analyst review, and step-up verification using only 1,074 analyst slots. We combine adaptive prediction sets (APS) from conformal prediction with shadow pricing from the bandits-with-knapsacks framework to ration analyst capacity dynamically. On the IEEE-CIS benchmark, our system reduces deployment risk by 43% over model alone and cuts cost by 52% over naive confidence thresholding. An intraday crossover at $\approx 40\%$ of the day elapsed where the system automatically shifts from analyst-heavy to verification-heavy routing demonstrates a dynamic that thresholding cannot replicate.

1 Introduction

Fraud detection is a human-AI collaboration problem: fully automated systems miss subtle fraud while fully manual review is operationally impossible. The deployment unit is a *triage policy* routing each transaction across automated approval, automated blocking, human analyst review, or step-up customer verification, a direct instance of the canonical rule of [1]:

$$\pi^{\star}(x) = \arg\min_{a \in \mathcal{A}} \; r_a(x;\, h) + c(a;\, x), \tag{1}$$

where $r_a(x; h)$ is conditional risk under action a and human h, and $c(a; x)$ is deployment cost. Naive confidence thresholding fails because it escalates uniformly into the uncertain zone regardless of budget: easy morning cases exhaust analyst capacity before harder afternoon cases arrive. We address this with **(1)** APS conformal prediction sets [2] as a coverage-guaranteed routing signal; and **(2)** shadow pricing [5] to ration analyst capacity dynamically, producing strict improvements over thresholding on every deployment metric.

2 Setup

2.1 Dataset and base model

We evaluate on the IEEE-CIS Fraud Detection dataset [11] (590,540 real card transactions, 3.5% fraud rate); full preprocessing details are in Appendix A. A LightGBM classifier [6] achieves validation AUC $= 0.9522$; we use AUC throughout as accuracy is uninformative under severe class imbalance.

2.2 Calibration

We apply temperature scaling [4] on the validation set (derivation in Appendix B.2), learning $T = 0.576$ and reducing ECE from 0.126 to 0.079. Residual miscalibration remains at high scores transactions assigned $\approx 70\%$ fraud probability show empirical fraud rates near 30% motivating conformal sets as the routing signal rather than raw softmax confidence.

2.3 Decision-theoretic framing

The action space is $\mathcal{A} = \{\texttt{auto\_approve}, \texttt{auto\_block}, \texttt{escalate}, \texttt{verify}\}$ with costs $c_{\text{auto}}=0.1$, $c_{\text{verify}}=2.0$, and $c_{\text{escalate}}=10.0$. We simulate one operational day: 10,000 transactions, daily budget $B=1{,}074$ analyst slots (40% of the conformal uncertain zone), and a morning/afternoon arrival ordering reflecting realistic intraday fraud patterns.

3 Method

3.1 System workflow

Figure 1 shows the full triage pipeline. Every incoming transaction is scored by the LightGBM classifier. Hard threshold zones resolve 73% of daily volume instantly: $p<0.05$ (68.8%) auto-approved; $p>0.80$ (4.6%) auto-blocked. The uncertain middle band feeds the shadow-price gate, which routes to a human analyst or step-up verification depending on remaining budget.

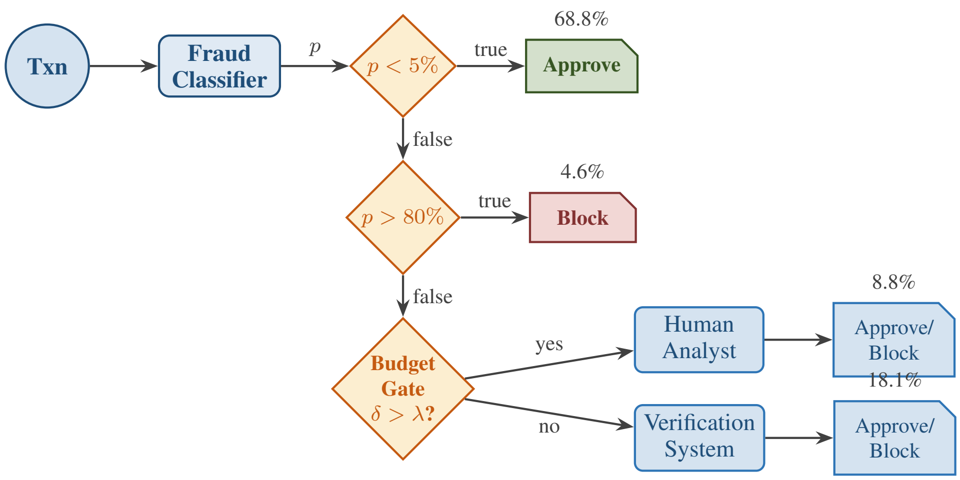

Figure 1: Fraud triage workflow. Hard threshold zones resolve 73% of daily volume automatically. The uncertain middle band ($5\% \leq p \leq 80\%$) feeds the budget-aware shadow-price gate: escalate to a human analyst if complementarity gap δ_{comp} exceeds the current shadow price λ; otherwise route to step-up verification. Percentages show daily routing volumes.

3.2 Conformal prediction sets as routing signal

Raw softmax confidence is an unreliable routing signal due to residual miscalibration. We use APS [2]: for each calibration point, the nonconformity score is the cumulative softmax mass required to include the true class (Eq. A2 in appendix). The conformal threshold at $\alpha=0.30$ yields $\hat{\tau}=0.9735$ with empirical coverage $0.9958 \geq 0.70$. The key finding: the auto-decide rate in the uncertain middle band is exactly 0% conformal prediction certifies every transaction between 5% 80% is genuinely uncertain rather than falsely precise.

3.3 Capacity-aware shadow pricing

Within the uncertain zone, we introduce a complementarity gap proxy $\delta_{\text{comp}}(x) = 1 - |2p(x) - 1|$, normalised to $[0, 1]$, which peaks at $p=0.5$ where human judgment adds most value. A shadow price λ initialised at 0 rises by $1/B$ after each escalation (dual ascent; derivation in Appendix B.3). The routing rule (Lecture 3, Eq. 3.5): escalate if $\delta_{\text{comp}}(x_t) > \lambda_t$ and budget remains; else verify. As λ rises, only the highest-complementarity cases justify analyst slots the policy is auditable: at any moment λ states the minimum complementarity gap required for escalation. Analysts are simulated at 85% accuracy; verification catches fraud at 60% (validated in Appendix D).

3.4 Baselines

S1 — model alone: automated at p=0.5, no humans.

S2 — confidence threshold: same hard zones, all middle-band cases escalated regardless of budget.

S3 — capacity-aware (ours): conformal routing with shadow pricing, B=1,074.

4 Experiments

4.1 Main results

Table 1: Deployment comparison on a simulated day of 10,000 transactions (346 fraud, 3.5%). Capacity-aware achieves lowest risk and FPR at half the cost of confidence thresholding. Higher FNR is the honest cost of finite capacity: verification catches 60% of fraud vs. the analyst's 85%.

Metric	S1: Model	S2: Threshold	S3: Capacity-aware
Risk (error rate)	0.0655	0.0641	**0.0371**
False negative rate	0.199	**0.095**	0.116
False positive rate	0.061	0.063	**0.034**
Escalation rate	0.0%	26.8%	**8.8%**
Verification rate	0.0%	0.0%	18.1%
Total cost	1,000	27,582	**13,142**
Cost per correct	0.107	2.947	**1.365**
Budget used	—	—	880/1,074 (82%)

Capacity-aware routing reduces error rate by 43% over S1 and total cost by 52% over S2. False positive rate of 0.034 is best across all strategies. The honest tradeoff: FNR of 0.116 exceeds S2's 0.095 because budget pressure routes cases to verification (60% catch) rather than analysts (85% catch) a cost the routing framework makes explicit.

4.2 Risk-coverage curve

Figure 2 confirms that S3 dominates S2 on the risk-coverage frontier. S1 maximises coverage at highest risk; S2 reduces risk by escalating broadly at enormous cost; S3 occupies the frontier with lower risk than S1 and lower cost than S2. The crossover in Appendix Figure 3 where cumulative verifications overtake escalations at $\approx 40\%$ of day explains *how*: shadow pricing automatically shifts routing as budget tightens, a dynamic confidence thresholding cannot produce.

5 Discussion

Limitations. Analysts are simulated at fixed 85% accuracy; real performance varies with fatigue and case complexity. The stationary reward assumption ignores skill atrophy and over-reliance dynamics documented in [1].

Transparency. A conformal set of size 2 has a precise meaning: the model cannot distinguish fraud from legitimate at the required coverage level more interpretable than a raw softmax score. The shadow price λ is inspectable at any moment, yielding an operationally auditable policy. SHAP values [8] can further identify which features drive individual scores [9].

Conclusion. Deployment constraints, not just model uncertainty must be first-class considerations in human-AI collaboration. Capacity-aware routing achieves 43% lower risk and 52% lower cost than baselines, and the intraday crossover emerges from the dual-ascent dynamics without manual re-parameterisation.

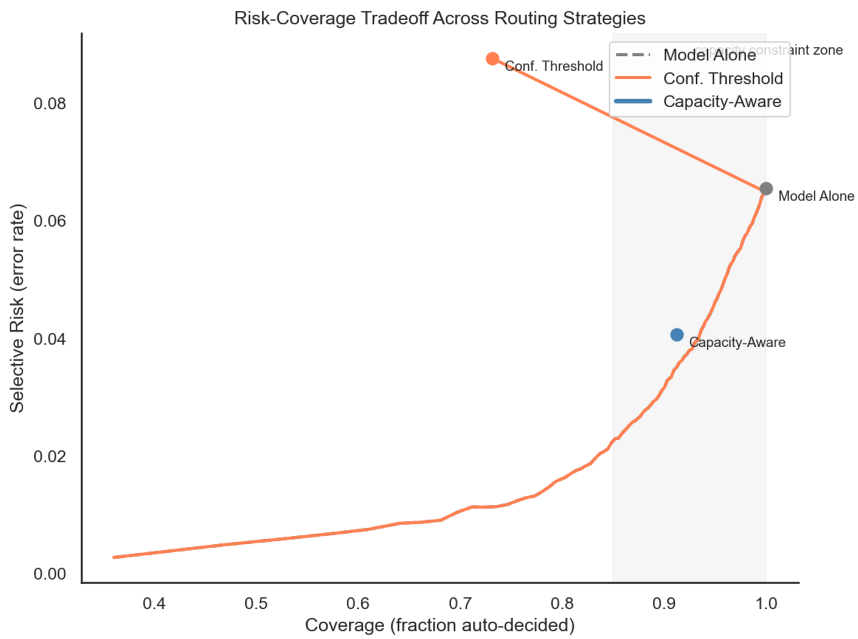

Figure 2: Risk-coverage tradeoff. The capacity-aware point (blue) lies below the confidence-threshold curve at similar coverage, achieving lower selective risk for the same automation rate. The shaded region marks coverage $> 85\%$. Full intraday dynamics (shadow price trajectory and the analyst-to-verification crossover at $\approx 40\%$ of day) are in Appendix Fig. 3.

Acknowledgments

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A Dataset and Base Model Details

The IEEE-CIS Fraud Detection dataset [11] contains 590,540 card transactions with 20,663 fraud cases (3.5% fraud rate). Transaction and identity tables are left-merged on `TransactionID`. After dropping columns with more than 50% missing values and imputing remaining missing values by column median (numeric) or a constant `missing` token (categorical), the resulting design matrix contains approximately 400 features covering card type (`card4`, `card6`), product category (`ProductCD`), transaction amount, device fingerprint, and behavioural pattern features.

We use a four-way stratified split: 60% train, 15% validation, 15% calibration, 10% test. The calibration set is kept strictly separate from validation because conformal prediction requires held-out data never used for model fitting or threshold selection; re-using these sets would invalidate the exchangeability assumption underlying the finite-sample coverage guarantee.

The LightGBM classifier [6] is trained with `n_estimators` $= 500$, `learning_rate` $= 0.05$, `num_leaves` $= 64$, and `scale_pos_weight` $= 27$ (the empirical negative-to-positive ratio, essential for handling class imbalance). Early stopping is performed on validation AUC with `first_metric_only` $=$ `True` to prevent the metric conflict that arises when binary log-loss is co-monitored on imbalanced data: log-loss degrades immediately as the model learns to push fraud scores higher, even while AUC continues to improve. The model trains for the full 500 iterations and achieves validation AUC $= 0.9522$.

B Formula Derivations

B.1 APS Nonconformity Scores

In the binary fraud setting, each transaction x receives two class probabilities: $\hat{p}_0(x) = 1 - p$ (legitimate) and $\hat{p}_1(x) = p$ (fraud). Sorted in descending order, the higher-scored class appears first.

For a calibration point with true label $y_i \in \{0, 1\}$, the APS nonconformity score is:

$$e_i = \sum_{j=1}^{k(x_i, y_i)} \hat{p}_{(j)}(x_i), \tag{2}$$

where $k(x_i, y_i)$ is the rank of the true class in the sorted order. This means:

- If $y_i = 1$ (fraud) and $p > 0.5$: fraud is ranked first, so $e_i = p$ — a small score, easy to include.
- If $y_i = 1$ (fraud) and $p < 0.5$: legitimate is ranked first, so $e_i = (1 - p) + p = 1$ — the model had to spend all its mass before reaching the true class.
- If $y_i = 0$ (legitimate) and $p < 0.5$: legitimate is ranked first, so $e_i = 1 - p$.
- If $y_i = 0$ (legitimate) and $p > 0.5$: fraud is ranked first, so $e_i = p + (1 - p) = 1$.

A well-classified example has a small nonconformity score; a misclassified or uncertain example has a score close to 1. The conformal threshold $\hat{\tau}$ (Equation **??** in the main paper) is then set so that the true class falls within the prediction set for at least $1 - \alpha$ of test transactions.

B.2 Temperature Scaling

Given raw model probabilities $p(x)$ and true labels y, temperature scaling learns a scalar $T > 0$ by minimising the negative log-likelihood on the held-out validation set:

$$T^\star = \arg\min_{T>0} \; -\frac{1}{n} \sum_{i=1}^{n} \Big[y_i \log \tilde{p}_i(T) + (1 - y_i) \log(1 - \tilde{p}_i(T)) \Big],$$

where $\tilde{p}_i(T) = \sigma\big(\mathrm{logit}(p_i)/T\big)$ and σ is the sigmoid function.

For $T < 1$, dividing the logit by $T < 1$ *increases* its magnitude, sharpening probabilities away from 0.5. For our model, $T = 0.576$: the raw model was assigning insufficiently extreme probabilities in the low-fraud-probability region where the bulk of legitimate transactions concentrate. Temperature scaling corrects this, reducing ECE from 0.126 to 0.079.

Note that $T < 1$ is often described as "overconfident" in the multi-class literature, but in the binary imbalanced setting the interpretation depends on which direction the miscalibration falls. Here the model was under-confident at low probabilities (legitimate transactions scored too close to the boundary), so $T < 1$ is the correct correction.

B.3 Shadow Pricing and Dual Ascent

The capacity-constrained routing problem can be written as a primal optimisation:

$$\min_{\pi} \mathbb{E}\big[r_{\pi(x)}(x;\, h)\big] \quad \text{subject to} \quad \mathbb{E}\big[d(\pi(x))\big] \leq B/N,$$

where $d(a) = \mathbf{1}[a = \texttt{escalate}]$ is the budget consumed and B/N is the per-transaction budget fraction. Forming the Lagrangian with dual variable $\lambda \geq 0$:

$$\mathcal{L}(\pi, \lambda) = \mathbb{E}\big[r_{\pi(x)}(x;\, h)\big] + \lambda\big(\mathbb{E}[d(\pi(x))] - B/N\big).$$

For a fixed λ, the optimal routing rule is pointwise:

$$\pi^{\star}_{\lambda}(x) = \arg\min_{a \in \mathcal{A}} \hat{r}_a(x;\, h) + \lambda\, d(a),$$

which is exactly Equation **??** in the main paper. Dual ascent updates λ after each escalation: $\lambda \leftarrow \lambda + \eta\,(d(\pi(x)) - B/N)$. Setting $\eta = 1/B$ gives the increment $1/B$ per escalation, so λ rises from 0 to 1 as all B budget slots are consumed.

B.4 Cost-Ratio Threshold

From Bhatt et al. [1], the Bayes-optimal two-action routing rule (automate vs. escalate) gives:

$$\pi^{\star}(x) = \texttt{escalate} \iff (1 - s(x))\, C_{\text{err}} > C_{\text{esc}},$$

which rearranges to the cost-ratio threshold:

$$\tau^{\star} = 1 - \frac{C_{\text{esc}}}{C_{\text{err}}}.$$

With our cost structure $C_{\text{esc}} = 10.0$ and $C_{\text{err}} = 1.0$ (normalising error cost to 1), this gives $\tau^{\star} = 1 - 10 = -9$, meaning the model should always escalate under this structure. In practice C_{err} must reflect the true cost of a wrong automated decision. Setting $C_{\text{err}} = 100$ (a missed fraud costs 100 units) gives $\tau^{\star} = 1 - 10/100 = 0.90$: escalate whenever the model is less than 90% confident. This illustrates that the threshold is a property of the deployment context, not the model.

C Conformal Alpha Selection

We evaluated two miscoverage levels. At $\alpha = 0.10$ (90% coverage guarantee), the conformal threshold is $\hat{\tau} = 0.9952$ and 63.7% of test transactions receive prediction set size 2. This is operationally impractical: escalating two thirds of daily transaction volume to human review would immediately exhaust any realistic analyst budget.

At $\alpha = 0.30$ (70% coverage guarantee), the threshold falls to $\hat{\tau} = 0.9735$ and 37.7% receive set size 2. After applying the hard threshold zones ($p < 0.05$ auto-approve, $p > 0.80$ auto-block), the effective uncertain zone that the routing policy must handle is 26.6% of daily volume a workable operating point for a team of analysts.

The choice of $\alpha = 0.30$ is therefore a deployment decision rather than a statistical one: it sets the coverage level at which the uncertain zone is large enough to be meaningful but small enough to be manageable. Empirical coverage on the test set at $\alpha = 0.30$ is 0.9958, substantially above the 0.70 guarantee, reflecting conservative calibration from the large held-out calibration set.

The conformal routing signal construction for both alpha values is visualised in Appendix Figure 6.

Table 2: Analyst simulation verification. Observed rates closely match their intended targets, confirming the simulation is functioning correctly.

Quantity	Target	Observed
Analyst error rate (Strategy 2)	0.15	0.155
Analyst error rate (Strategy 3)	0.15	0.147
Verification fraud catch rate	0.60	0.628
Fraud cases correctly blocked (Strategy 3)	—	306 / 346 (88.4%)

D Analyst Simulation Verification

To verify that the analyst simulation behaved as intended, we ran a diagnostic after Section 4 of the experiment notebook. Table 2 reports the observed rates against their targets.

The 306/346 fraud detection figure (88.4%) reflects the combined contribution of auto-block (high-confidence cases), analyst escalation, and step-up verification. The remaining 40 missed fraud cases are transactions in the uncertain zone that were routed to verification and not caught by the 60% catch rate.

E Supporting Figures

E.1 Intraday Capacity Dynamics

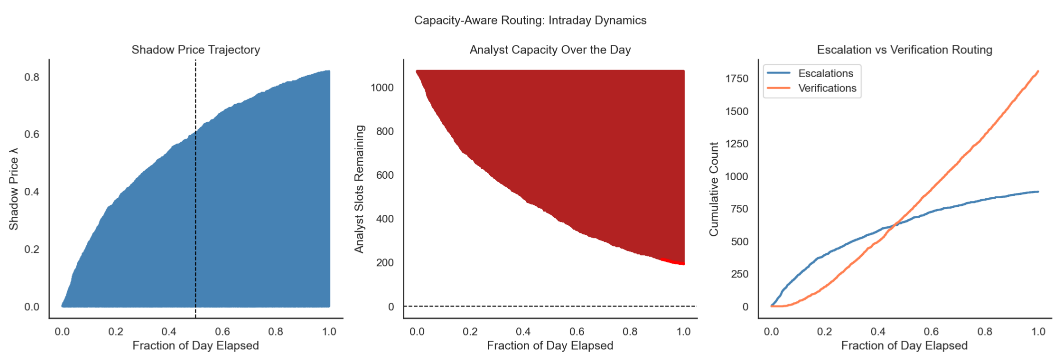

Figure 3: Intraday dynamics of capacity-aware routing. *Left*: shadow price λ rises from 0 to 0.82 as analyst budget depletes, implementing the dual-ascent update $\lambda \leftarrow \lambda + 1/B$ after each escalation. *Centre*: analyst slots remaining over the working day, depleting steadily through the afternoon. *Right*: cumulative escalations (blue) and verifications (coral) showing a crossover at $\approx 40\%$ of day elapsed, where the system transitions from analyst-heavy to verification-heavy routing as the shadow price rises above the complementarity gap of most incoming transactions.

E.2 Model Calibration

E.3 Conformal Routing Signal Construction

E.4 Strategy Comparison Dashboard

E.5 Dataset Overview

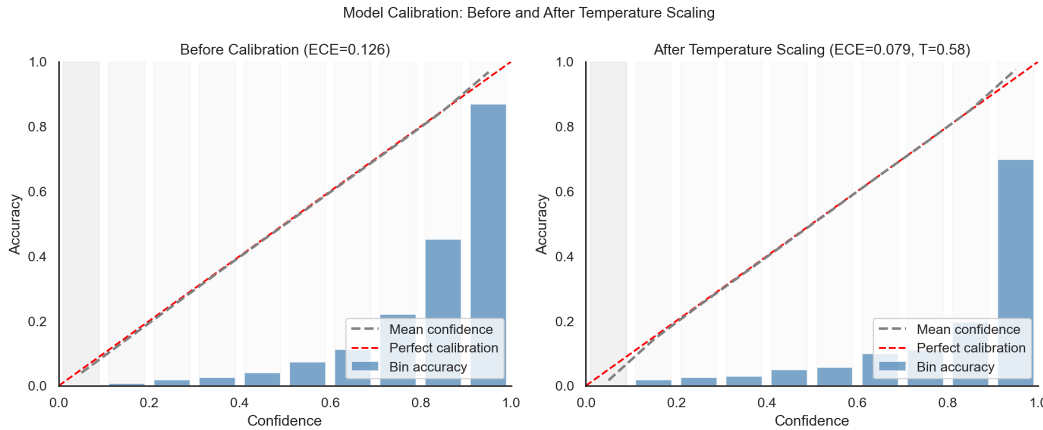

Figure 4: Reliability diagrams before and after temperature scaling (equal-mass binning, 10 bins). ECE falls from 0.126 to 0.079. The visible bar at high confidence ($\approx$0.70) shows accuracy $\approx$0.30 before scaling, confirming the model is overconfident in the high-score region. Most bins concentrate near zero because 90% of transactions receive fraud probability below 0.10 — a consequence of the 3.5% fraud rate rather than a defect of the diagram. The residual miscalibration at high scores motivates using conformal prediction sets as the routing signal.

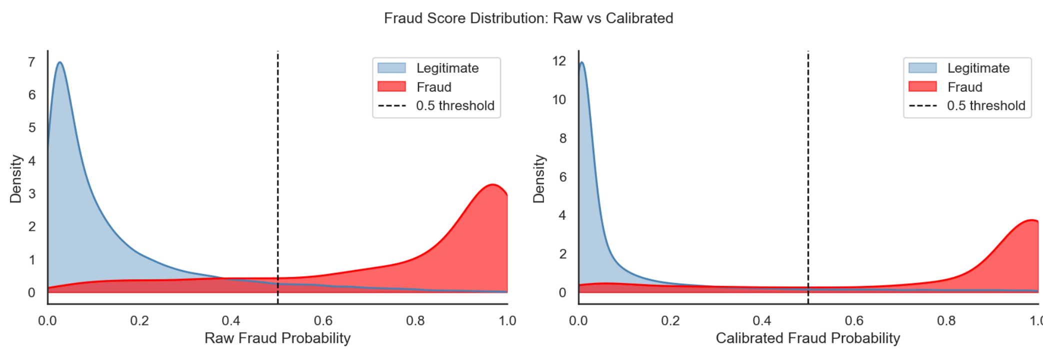

Figure 5: Fraud score distributions before and after temperature scaling. The overlap region between 0.05 and 0.15 is the ambiguous zone where the routing policy adds most value. The 0.5 threshold catches almost nothing because the model rarely assigns such high fraud probability, illustrating why naive thresholding at 0.5 is unsuitable for this dataset.

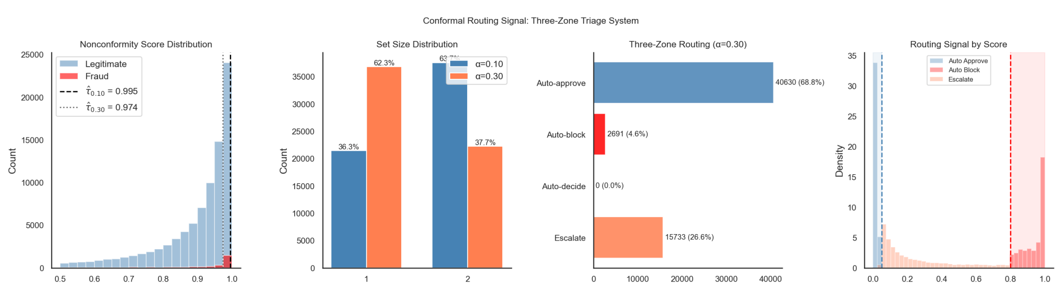

Figure 6: Conformal routing signal construction. *Left*: nonconformity score distributions by true class, with dashed lines showing thresholds at $\alpha = 0.10$ and $\alpha = 0.30$. *Centre-left*: set-size distributions at both operating points, showing the sensitivity to α. *Centre-right*: three-zone routing distribution at $\alpha = 0.30$ — 68.8% auto-approve, 4.6% auto-block, 26.6% escalation zone, 0% auto-decide in the middle band. *Right*: fraud probability distribution coloured by routing zone.

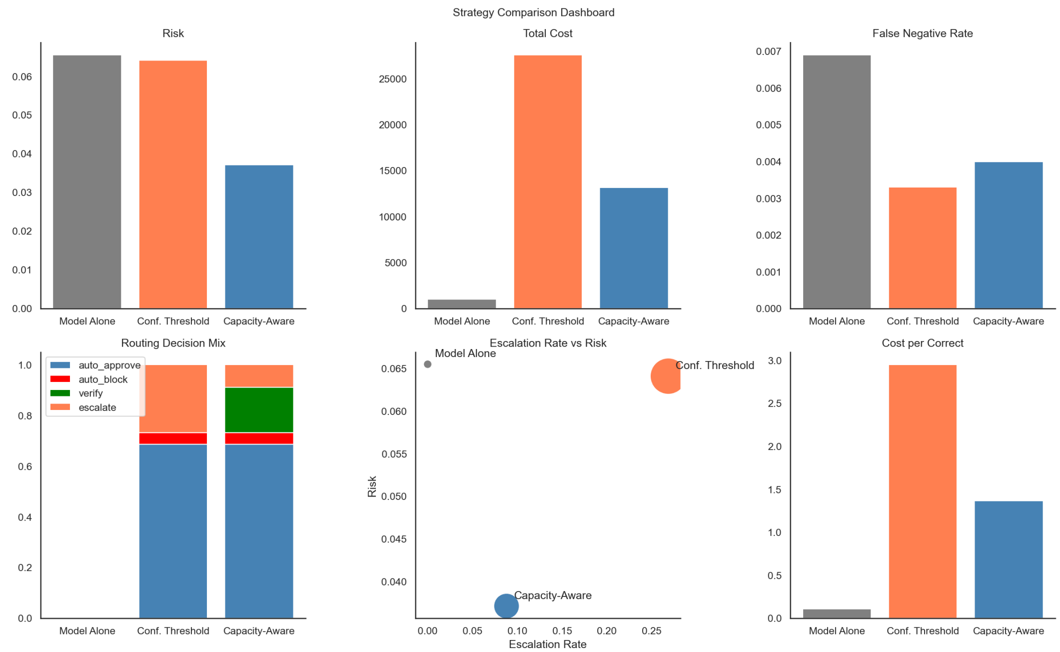

Figure 7: Full strategy comparison dashboard. *Top row*: risk, total cost, and false negative rate per strategy. *Bottom row*: routing decision mix (stacked bars), escalation rate vs. risk scatter with point size proportional to total cost, and cost per correct decision. The scatter plot summarises the core result: capacity-aware routing (blue) achieves both lower risk and lower escalation rate than confidence thresholding (coral) at intermediate cost.

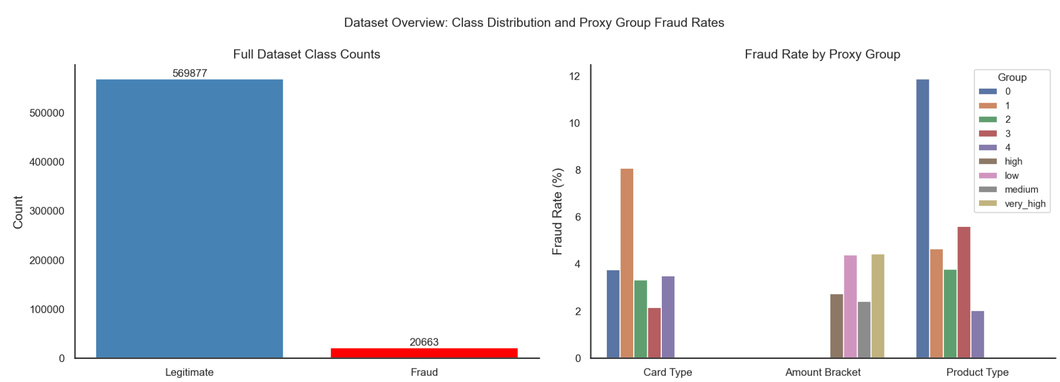

Figure 8: Dataset overview. *Left*: class imbalance — 569,877 legitimate versus 20,663 fraud transactions (3.5% fraud rate). *Right*: fraud rate by proxy group (`card4`, amount bracket, `ProductCD`), showing substantial variation especially across product types (2%–12%).